MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIFTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG5DSCCND301 – PEDIATRIC NUTRITION

Duration: 2 hrs Maximum Marks: 70

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Each question is tagged with the appropriate Bloom's Revised Taxonomy level (Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating.) and the relevant course outcome (CO) number.

* Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S).

Part A
Objective Questions
Answer **All** Questions
Each question carries **1** mark

1. The birth weight cut off for low birth weight (LBW) is ……. [U] [2]
   A) <3.5 Kg
   B) <2.5 Kg
   C) <1.5 Kg
   D) <2 Kg

2. State the full form of IUGR. [K] [2]
   A) Intrauterine Growth Restriction
   B) Intrauterine Growth Response
   C) Intrauterine Growth Rate
   D) Intrauterine Growth Regulation

3. Name the growth chart recommended by IAP for children below 5 years of age: [U] [1]
   A) NCHS growth chart
   B) WHO growth chart
   C) CDC growth chart
   D) Gomez growth chart

4. Explain the primary purpose of the Pediatric Assessment Triangle (PAT). [U] [1]
   A) To assess the nutritional status of a child
   B) To identify critically ill or injured children rapidly

C) To determine the developmental age of a child
D) To assess the immunization status of a child

5. Identify the two primary disorders that comprise inflammatory bowel disease (IBD). [U] [4]
   A) Celiac disease and Crohn's disease
   B) Crohn's disease and ulcerative colitis
   C) Ulcerative colitis and irritable bowel syndrome
   D) Crohn's disease and gastroesophageal reflux disease

6. Identify the investigation used to record abnormal electrical activity in the brain of a child with epilepsy. [U] [4]
   A) MRI
   B) EEG
   C) CT scan
   D) X-ray

7. Identify the gene mutated in cystic fibrosis. [U] [4]
   A) HBB gene
   B) CFTR gene
   C) BRCA1 gene
   D) F8 gene

8. A child is advised to choose a beverage with very low cariogenic potential. Which beverage should be selected? [U] [3]
   A) Fruit juice
   B) Carbonated drink
   C) Plain milk
   D) Sweetened milk drink

9. A child develops symptoms within minutes of consuming a food, with urticaria and respiratory difficulty. Which type of food allergy is most consistent with this presentation? [U] [3]
   A) IgE- mediated allergy
   B) Non-IgE-mediated allergy
   C) Food intolerance
   D) Enzyme deficiency

10. Identify the clinical condition that may develop as a long-term consequence of persistent undernutrition in children. [U] [3]
   A) Failure to thrive
   B) Accelerated growth
   C) Increased muscle mass
   D) Early maturation

[1 x 10 = 10 marks]

Part B

Short Answer Type Questions
Answer any **5** Questions
Each question carries **2** marks

11. Define Type 1 Diabetes Mellitus (T1DM) and state any one of the underlying causes of insulin deficiency. [K] [4]
12. Describe Modified Atkins Diet (MAD)? [U] [4]
13. Identify the anthropometric measurement used to assess body fatness and explain how it is used. [U] [1]
14. Explain lactose intolerance and describe its different types? [U] [2]
15. Identify the enteral feeding method that may be used when oral feeding is inadequate in children with cerebral palsy. [K] [2]
16. Explain any two causes of childhood obesity ? [U] [3]
17. Define food allergy. Mention any two risk factors for food allergy. [K] [3]

[2 x 5 = 10 marks]

Part C

Short Essay Type Questions
Answer any **6** Questions
Each question carries **5** marks

18. Explain insulin therapy and its types and components. [A] [4]
19. Explain cystic fibrosis and describe its symptoms and clinical features? [A] [4]
20. Explain Pediatric Assessment Triangle and describe its core components? [A] [1]
21. Compare the 24-hour recall, Food Frequency Questionnaire, and food record methods used for dietary assessment in children. [An] [1]
22. Explain the nutritional requirements and dietary management of a child with celiac disease. [U] [2]
23. Explain the feeding challenges faced by children with developmental disabilities. [U] [2]
24. Explain the types of dental caries in children. [U] [3]
25. Describe the short-term consequences of underweight. [U] [3]

[5 x 6 =30 marks]

Part D

Essay Type Questions

Answer any **2** Questions

Each question carries **10** marks

26. Explain how you would develop a comprehensive monitoring and documentation plan for a child with epilepsy? [A] [4]
27. Explain the procedure for recording and plotting a child's anthropometric measurements on the WHO Growth Chart and discuss how the plotted growth curve is interpreted. [A] [1]
28. Analyze the nutritional considerations in children with congenital heart disease (CHD) and formulate appropriate dietary management strategies to meet their increased nutritional requirements. [U] [2]
29. Explain the long term consequences of undernutrition. [A] [3]

[10 x 2 =20 marks]